Avcorr - instruction

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Description

Parallelized and multi-functional code for computing area- or space- averaged correlation functions $\bar{\xi}_J$ of orders 2nd-9th, along with cumulants s_J, as functions of scale(s).

It allows for the calculation in four different modes (VERSIONs):

- **angular**: angular counts in $[RA, DEC]$ catalog
- **BOX**: count in spheres within cubic $[X, Y, Z]$ simulation box
- **BOX_ellipses**: counts in ellipsoids within cubic $[X, Y, Z]$ simulation box
- **LC_ellipses**: counts in ellipsoids within $3D[X, Y, Z]$ lightcone

for more details check section about code versions.

1 Requirements, compilation and running

Requirements

- c++17 or newer
- mpi.h library for c++
- libhdf5-cpp

To compile the code, type: **make**
To run, type (X stands for the number of desired parallel jobs):

Running

mpiexec -np X ./Avcorr.exe

Or use bash script *./run* by typing: **./run X**
firstly ensuring if it has permissions (chmod +700 *./run*)

- **Computations can be restarted in any moment if user decided to stop it. The restarted code will just ignore the existing counts.**
- **Input data files need to have the same extension and be located in the same directory as specified in parameter file.**
- **Output scales are always spaced logarithmically**

2 Important files and directories

- Data/ - default location of data for which code calculates the statistics (see description of Datafiles entry in parameter file)
- Results/ - stores results after the run
- Randoms/ - contains optional random catalogs
  (valid if Random_provided=1, see parameter file section)

3 Different code versions, inputs and outputs

angular

Standard counts-in-cells(CiC), counts within circles located on $[RA, DEC]$ sphere.
Input file format: RA, DEC in degrees (datafiles either ASCII or HDF5, see parameter file).

Output files:
Results/[fname]_merrout.txt (contains $\bar{\xi}_J$ results):
θ, $\bar{\xi}_2$, $u_{\bar{\xi}_2}$, $\bar{\xi}_3$, $u_{\bar{\xi}_3}$, $\bar{\xi}_4$, $u_{\bar{\xi}_4}$, $\bar{\xi}_5$, $u_{\bar{\xi}_5}$, $\bar{\xi}_6$, $u_{\bar{\xi}_6}$, $\bar{\xi}_7$, $u_{\bar{\xi}_7}$, $\bar{\xi}_8$, $u_{\bar{\xi}_8}$, $\bar{\xi}_9$, $u_{\bar{\xi}_9}$,

Results/[fname]_merrout_Sn.txt (contains s_J results):
θ, 1, 0, s_3, u_{s_3}, s_4, u_{s_4}, s_5, u_{s_5}, s_6, u_{s_6}, s_7, u_{s_7}, s_8, u_{s_8}, s_9, u_{s_9},

where:
fname - corresponding input datafile without extension (or set of input files, check *Combine_reals* and *Real_template* parameters)
θ - angular scale in degrees, defined by parameters $Rmin, Rmax, nR$ (check parameters section)
$\bar{\xi}_J/s_J$ - J-th order averaged correlated function/hierarchical amplitude
$u_{\bar{\xi}_J}/u_{s_J}$ - error of $\bar{\xi}_J/s_J$, check error calculation section. The 1,0 values at 2nd and 3rd column of _merrout_Sn files stand for consistency of the format since $s_2 = 1$ by definition.
Output files will have additional columns if *ErrPoisson* parameter is set to 1 - check *ErrPoisson* description.

BOX

Counts within spheres in BOX (each axis in $[0, boxsize]$ range)
Input file format: X, Y, Z, units the same as the units of sphere sizes (datafiles either ASCII or HDF5, see parameter file).

Output files:
Results/[fname]_merrout.txt (contains $\bar{\xi}_J$ results):
R, $\bar{\xi}_2$, $u_{\bar{\xi}_2}$, $\bar{\xi}_3$, $u_{\bar{\xi}_3}$, $\bar{\xi}_4$, $u_{\bar{\xi}_4}$, $\bar{\xi}_5$, $u_{\bar{\xi}_5}$, $\bar{\xi}_6$, $u_{\bar{\xi}_6}$, $\bar{\xi}_7$, $u_{\bar{\xi}_7}$, $\bar{\xi}_8$, $u_{\bar{\xi}_8}$, $\bar{\xi}_9$, $u_{\bar{\xi}_9}$,

Results/[fname]_merrout_Sn.txt (contains s_J results):
R, 1, 0, s_3, u_{s_3}, s_4, u_{s_4}, s_5, u_{s_5}, s_6, u_{s_6}, s_7, u_{s_7}, s_8, u_{s_8}, s_9, u_{s_9},

where:
fname - corresponding input datafile without extension (or set of input files, check *Combine_reals* and *Real_template* parameters)
R - scale in the same units as in catalog, defined by parameters $Rmin, Rmax, nR$ (check parameters section)
$\bar{\xi}_J/s_J$ - J-th order averaged correlated function/hierarchical amplitude
$u_{\bar{\xi}_J}/u_{s_J}$ - error of $\bar{\xi}_J/s_J$, check error calculation section. The 1,0 values at 2nd and 3rd column of _merrout_Sn files stand for consistency of the format since $s_2 = 1$ by definition.
Output files will have additional columns if *ErrPoisson* parameter is set to 1 - check *ErrPoisson* description.

BOX_ellipses

Counts within ellipses in BOX (each axis in [0,boxsize] range). Ellipses have axes: axa along X axis, axb along both Y and Z axes. The code creates a grid of axa and axb values with ranges defined in parameter file and for each set of [axa,axb] computes the statistics. Option useful for investigating catalogs with RSD along X axis.
Input file format: X, Y, Z, units the same as the units of sphere sizes (datafiles either ASCII or HDF5, see parameter file).

Output files:
Results/[fname]_merrout.txt (contains $\bar{\xi}_J$ results):
axa, axb, $\bar{\xi}_2$, $u_{\bar{\xi}_2}$, $\bar{\xi}_3$, $u_{\bar{\xi}_3}$, $\bar{\xi}_4$, $u_{\bar{\xi}_4}$, $\bar{\xi}_5$, $u_{\bar{\xi}_5}$, $\bar{\xi}_6$, $u_{\bar{\xi}_6}$, $\bar{\xi}_7$, $u_{\bar{\xi}_7}$, $\bar{\xi}_8$, $u_{\bar{\xi}_8}$, $\bar{\xi}_9$, $u_{\bar{\xi}_9}$,

Results/[fname]_merrout_Sn.txt (contains s_J results):
axa, axb, 1, 0, s_3, u_{s_3}, s_4, u_{s_4}, s_5, u_{s_5}, s_6, u_{s_6}, s_7, u_{s_7}, s_8, u_{s_8}, s_9, u_{s_9},

where:
fname - corresponding input datafile without extension (or set of input files, check *Combine_reals* and *Real_template* parameters)
axa, axb - semi axes of ellipsoid, parallel and perpendicular to line-of-sight (LOS)
$\bar{\xi}_J/s_J$ - J-th order averaged correlated function/hierarchical amplitude
$u_{\bar{\xi}_J}/u_{s_J}$ - error of $\bar{\xi}_J/s_J$, check error calculation section. The 1,0 values at 3rd and 4th column of _merrout_Sn files stand for consistency of the format since $s_2 = 1$ by definition.
Output files will have additional columns if *ErrPoisson* parameter is set to 1 - check *ErrPoisson* description.

LC_ellipses

Counts within line-of-sight-elongated ellipses in 3D catalog made from lightcone. Ellipses have axes: axa along line-of-sight(LOS), axb along two transverse directions. The code creates a grid of axa and axb values with ranges defined in parameter file and for each set of [axa,axb] computes the statistics. Option useful for investigating higher-order statistics in 3D lightcones including RSD effects.
Input file format: X, Y, Z, units the same as the units of sphere sizes (datafiles either ASCII or HDF5, see parameter file).

Output files:
Results/[fname]_merrout.txt (contains $\bar{\xi}_J$ results):
axa, axb, $\bar{\xi}_2$, $u_{\bar{\xi}_2}$, $\bar{\xi}_3$, $u_{\bar{\xi}_3}$, $\bar{\xi}_4$, $u_{\bar{\xi}_4}$, $\bar{\xi}_5$, $u_{\bar{\xi}_5}$, $\bar{\xi}_6$, $u_{\bar{\xi}_6}$, $\bar{\xi}_7$, $u_{\bar{\xi}_7}$, $\bar{\xi}_8$, $u_{\bar{\xi}_8}$, $\bar{\xi}_9$, $u_{\bar{\xi}_9}$,

Results/[fname]_merrout_Sn.txt (contains s_J results):
axa, axb, 1, 0, s_3, u_{s_3}, s_4, u_{s_4}, s_5, u_{s_5}, s_6, u_{s_6}, s_7, u_{s_7}, s_8, u_{s_8}, s_9, u_{s_9},

where:
fname - corresponding input datafile without extension (or set of input files, check *Combine_reals* and *Real_template* parameters)
axa, axb - semi axes of ellipsoid, parallel and perpendicular to line-of-sight (LOS)
$\bar{\xi}_J/s_J$ - J-th order averaged correlated function/hierarchical amplitude
$u_{\bar{\xi}_J}/u_{s_J}$ - error of $\bar{\xi}_J/s_J$, check error calculation section. The 1,0 values at 3rd and 4th column of _merrout_Sn files stand for consistency of the format since $s_2 = 1$ by definition.
Output files will have additional columns if *ErrPoisson* parameter is set to 1 - check *ErrPoisson* description.

Common output files (for every code mode (VERSION))

Output files with final results are described in text above, separately for each code version.

Results/[fname]_moments_nnR_X.fndhst
(contains histogram of counts from X-th scale):
N, n_c,
where N - number of objects within circle, n_c - number of circles with N objects.
Results/log.txt
(contains log of the run)

Results/*_moments_*.txt (temporary files with yet-unmerged results, will be removed at the end if $Clean = 1$ parameter is set)

4 Randoms

Random files (must be located in Randoms/ directory) used in the code are necessary only if $Random_provided = 1$ in parameter file. The format is exactly the same as data files for corresponding VERSION (version is set in parameter file). If $USE_HDF5 = 1$, random data needs to be within POS_DSET dataset, otherwise ($USE_HDF5 = 0$) random file(s) have to contain columns with positions only. Randoms define positions of circles/spheres/ellipses for making counts and are useful for data with irregular shape. It is crucial not to use randoms with positions exceeding the catalog ranges. Check description of parameters corresponding with randoms: Random_provided, Random_file, Random_proc, Random_fullr and Random_sigmac.

The code includes an option for processing randoms for irregular footprint (see Random_proc parameter description). Given full randoms, for every scale considered separately it prepares reduced randoms considering only positions where random density does not deviate down from average by more than Random_sigmac standard deviations. This procedure prevents counts from extending partially footprint due to the finite size of the counting cells.
[Attention]: reducing randoms for now available only for VERSION=angular

5 Parameter file

The param.txt file contains the code settings. For clarity it is divided into parts containing parameters working only for specific code VERSION (angular/BOX/ BOX_ellipses/LC_ellipses). The parameters are:

- **VERSION** - code mode: angular/BOX/ BOX_ellipses/LC_ellipses. Switches between the modes, based on this different data formats and outputs will be used.
- **USE_HDF5** - input data format: 0 - ASCII, 1 - HDF5
- **POS_DSET** - dataset with coordinates (applies only if only if USE_HDF5=1)
- **cols_pos** - columns with coordinates (applies only if only if USE_HDF5=0)

Common parameters (used for **every** VERSION):

- **nreals** - number of sub-probes for errors calculation. Counts are splitted into [nreals] parts, then for each sub-probe the code calculates the statistics. Output result is mean from sub-probes +/- one standard deviation. Check the section about error calculations.
- **kappa** - fraction of statistically independent information contained within one sub-probe, i.e. for $VERSION = BOX$ number of spheres considered for each sub-probe will be: $kappa \times boxsize^3/(4/3\pi R^3)$, where R is the sphere size. Use $kappa << 1$ to avoid error underestimation (otherwise every sub-probe will contain approximately the same information) and $kappa \times nreals > 1$ to extract full information. Check the section about error calculations.
- **Cmin** - minimum number of circles/spheres/ellipses for CiC
- **Cmax** - maximum number of circles/spheres/ellipses for CiC
- **Datafiles** - datafile names from separated by space/tab (with extension). Examples:
  - If Datafiles=*, the code selects all files in Data/ directory
  - If Datafiles=path/*, the code selects all files from path/ directory

  - If Datafiles=file1.h5 file2.h5, the code selects file1.h5 and file2.h5 from Data/ directory, analogically for specified path/
  - [Attention]: all files needs to have the same extension and be located in the same directory
  - [Attention]: if * was used firstly, only files corresponding with that entry will be considered, e.g. for Datafiles=path/* file2.dat, code will read only files in path/

- **Recalc** - [0/1] if code is restarted and counts already exist, recalculating the moments [1], or not [0] to save the time

- **Clean** - [0/1] cleaning unnecessary files at the end

- **ErrPoisson** - [0/1] switch whether we want Poisson errors calculation or not. For $ErrPoisson = 0$, there are no columns with Poisson errors in output file. If $ErrPoisson = 1$, the errors for orders 2-9 are added to the file, so the format is like:
R, $\bar{\xi}_2$, $u_{\bar{\xi}_2}$, $\bar{\xi}_3$, $u_{\bar{\xi}_3}$, $\bar{\xi}_4$, $u_{\bar{\xi}_4}$, $\bar{\xi}_5$, $u_{\bar{\xi}_5}$, $\bar{\xi}_6$, $u_{\bar{\xi}_6}$, $\bar{\xi}_7$, $u_{\bar{\xi}_7}$, $\bar{\xi}_8$, $u_{\bar{\xi}_8}$, $\bar{\xi}_9$, $u_{\bar{\xi}_9}$,
P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9,
where P_J is Poisson error for order J.

- **Combine_reals** - [0/1] switch whether our files are different realisations of one model (check Real_template parameter)

- **Real_template** - template telling how are realisations marked if $Combine_reals = 1$, e.g. if $Real_template = "box*_"$, and the data files are AAbox1_data, AAbox2_data, AAbox3_data, the code will combine them: the result will be AAboxN_data_merrout.txt: averaged over realisations and errors from standard deviation. One can have any number of files with more than one model, for example: Files: AAbox1_data, AAbox2_data, AAbox3_data, BBbox1_data, BBbox2_data, BBbox3_data, BBbox4_data, will be combined into AAboxN_data_merrout.txt (3 realisations included) and BBboxN_data_merrout.txt (4 realisations included), just Real_template must be the same.
If $Random_file = *$, the Randoms/ directory must still contain randoms for each file in Data/ separately, even if these are the same model but different realisations

- **Random_provided** - random file(s) provided? 0 - drawing random, 1 - reading random file (s) [check Random_file description].
Drawing randoms:

  - for $VERSION = angular$: random points within [RA,DEC] rectangle, not closer than current circle size to catalog borders
  - for $VERSION = BOX$: random points in $[R, boxsize - R]^3$ cube, where R - current sphere size
  - for $VERSION = BOX_ellipses$: random points in $[ax, boxsize - ax]$ ranges, where ax=axa for axis X and ax=axb for axes Y and Z, check parameters for BOX_ellipses and LC_ellipses
  - for $VERSION = LC_ellipses$: random points between spheres of radii DCMIN+axa and DCMAX-axa. [Attention:] for LC_ellipses random drawer ignores any angular cuts, it just draws them for all directions. If your catalog has angular cuts, use own randoms.

- **Random_file** - random file name (if specific file name imposed here, every datafile will be assigned to the same random file); if $Random_file = *$, code reads multiple randoms: for each datafile, random filename is assumed to be: Randoms/Randoms_[datafile] will be assigned to each [datafile]. For example, if $Random_file = *$ and Data/ contains

catalogs A.txt and B.txt, then Randoms/ directory has to contain Randoms_A.txt and Randoms_B.txt files. Random file(s) format for ASCII datafiles ($USE_HDF5 = 0$) is assumed to have only columns with coordinates, while for HDF5 datafiles ($USE_HDF5 = 1$) the positions are assumed to be within POS_DSET dataset.

- **Random_proc** - [0/1] processing randoms? 0 - using Random_file as positions for counts, 1 - creating reduced randoms separately for every scale to account for cells size effect (useful for irregular footprints, see Randoms section). [Attention]: reducing randoms for now available only for VERSION=angular

- **Random_fullr** - ranges where the data (or randoms) is continuous (no holes, just rectangular shape), for counting average randoms density and standard deviation used in random reducing method (applies only if Random_proc=1). For $VERSION = angular$, it has format:
ramin ramax decmin decmax
defining RA,DEC ranges.

- **Random_sigmac** - defines cut tolerance in random reducing: how many standard deviations counts at given point can maximally differ from random average, to consider this point as safe for CiC.

Parameters for $VERSION =$ **angular** only:

If $Random_file = *$ and the sky footprint is complicated, right-ascension and declination ranges in param.txt file have to exceed extreme values from catalogs - for pixelization purposes

- **ramin** - catalog right-ascension lower range in degrees
- **ramax** - catalog right-ascension upper range in degrees
- **decmin** - catalog declination lower range in degrees
- **decmax** - catalog declination upper range in degrees
- **Areaf** - catalog sky area in square degrees (for optimization purposes)

Parameters both for $VERSION =$ **angular** and **BOX**:

- **Rmin** - smallest scale considered
- **Rmax** - biggest scale considered
- **nR** - number of scales considered

Parameters both for $VERSION =$ **BOX** and **BOX_ellipses**:

- **Boxsize** - box size in the same units as random spheres/ellipses

Parameters both for $VERSION =$ **BOX_ellipses** and **LC_ellipses**:

- **axamin** - lower range of ellipses axes along X axis (VERSION=BOX_ellipses) / LOS (VERSION=LC_ellipses)
- **axamax** - upper range of ellipses axes along X axis (VERSION=BOX_ellipses) / LOS (VERSION=LC_ellipses)
- **axbmin** - lower range of ellipses axes perpendicular to X axis (VERSION=BOX_ellipses) / LOS (VERSION=LC_ellipses)

- **axbmax** - upper range of ellipses axes perpendicular to X axis (VERSION=BOX_ellipses) / LOS (VERSION=LC_ellipses)
- **naxa** - number of different axes along X/LOS considered - grid size
- **naxb** - number of different axes perpendicular to X/LOS considered - grid size

Parameters for $VERSION$ = **LC_ellipses** only:

- **DCMIN** - minimum comoving distance (from observer) considered, the same units as for ellipse axes
- **DCMAX** - minimum comoving distance (from observer) considered, the same units as for ellipse axes

6 Errors calculation

Combine_reals set to 0

Errors are being calculated by splitting full counts into [$nreals$] sub-probes (without returning) and computing standard deviation on the statistics calculated for each sub-probe. Additionally, the code provides independently Poisson errors assuming that $error = \sqrt{counts}$ and calculating the errors of $\bar{\xi}_J/s_J$ using error propagation.

Combine_reals set to 1

Combining results from files according to Real_template (check Parameter file section), computing average and standard deviation. The code finds independent models and number of realizations automatically.

7 Example runs and plots

The directory Examples/ contains example results for different configurations along with data and parameter file(s) necessary for recreating them. Each example also contains python plotting scripts for visualization. For more details, check Examples/[example]/ReadMe.txt file. Examples have been computed on data from $COLAVERSE$ simulation prepared by Prof. Wojciech Hellwing and described in Drozda et al. (2025).

8 Troubleshooting and future improvements

The code predicts potential errors in parameter file (mismatching number of columns, nonphysical values i.e. $Rmin < 0$, empty $Data/$ directory and many other). If the code breaks or unexpected behavior occurs:

- check $Results/log.txt$ file. In case of problems in param.txt it should report errors in lines starting with [Error]
- check if datafiles do not contain NaN, INF or other problems inside the data
- check if proper format is specified - code will break if USE_HDF5 does not match data type
- if ASCII data is used ($USE_HDF5 = 0$), check if file headers (lines starting with # are only at the beginning of files and are not separated by non - #- starting lines)

- check if data files have the same extension and are located in the same directory

Future improvements to add:

- periodicity in $VERSION = angular$
- generalize optimal pixelization into analytical expressions

9 Citation & references

The code uses equations from Gaztanaga (1994) to compute connected and shot-noise corrected moments of counts in cells.
While using this code, please cite Drozda et al. (2025) and refer to:
$https://github.com/Pawel-96/Avcorr$.

References

P. Drozda, W. A. Hellwing, and M. Bilicki. Anisotropic counts-in-cells in redshift space: A new route to cosmological constraints from galaxy surveys, 2025. URL https://arxiv.org/abs/2506.01762.

E. Gaztanaga. High-Order Galaxy Correlation Functions in the APM Galaxy Survey. *Monthly Notices of the Royal Astronomical Society*, 268:913, June 1994. doi: 10.1093/mnras/268.4.913.